

Question:

Assume that B Corporation has the following total profit function equation: Total Profit = $\pi = - \$3,000 - \$2,400 Q + \$350Q^2 - \$8.333Q^3$ a) Compute the marginal profit b) Determine its output quantities

Answer Choices:

- A. 22 and 7 units
- B. 21 and 8 units
- C. 23 and 6 units
- D. 26 and 2 units
- E. 24 and 4 units
- F. 25 and 3 units
- G. 20 and 5 units
- H. 27 and 1 unit
- I. 18 and 10 units
- J. 19 and 9 units

Reasoning + Answer:

</tool_call> reasoning process here

The total profit function is given by :

$$\pi = -3000 - 2400Q + 350Q^2 - 8.333Q^3$$

a) Marginal profit (MP) is the derivative of total profit with respect to Q :

$$MP = d\pi/dQ = d/dQ [-3000 - 2400Q + 350Q^2 - 8.333Q^3]$$

Differentiate term by term :

- d/dQ (-3000) = 0
- d/dQ (-2400Q) = -2400
- d/dQ (350Q^2) = 700Q
- d/dQ (-8.333Q^3) = -25Q^2 (since 8.333 Ã 3 = 25)

So :

$$MP = -2400 + 700Q - 25Q^2$$

b) To find the output quantity that maximizes profit , set marginal profit to zero :

$$MP = 0$$
$$\hat{= } -2400 + 700Q - 25Q^2 = 0$$

Multiply both sides by - 1 :

$$2400 - 700Q + 25Q^2 = 0$$

Divide entire equation by 25 to simplify :

$$96 - 28Q + Q^2 = 0$$
$$\hat{= } Q^2 - 28Q + 96 = 0$$

Solve the quadratic equation :

$$Q = [28 \hat{+ } \hat{- } (28^2 - 4 \hat{= } 1 \hat{= } 96)] / 2$$
$$= [28 \hat{+ } \hat{- } (784 - 384)] / 2$$
$$= [28 \hat{+ } \hat{- } 400] / 2$$
$$= [28 \hat{+ } 20] / 2$$

So :

$$Q = (28 + 20) / 2 = 48 / 2 = 24$$
$$Q = (28 - 20) / 2 = 8 / 2 = 4$$

Thus , the output quantities that maximize profit are Q = 24 and Q = 4 .

So the correct pair is 24 and 4 units .

Answer choice : E . 24 and 4 units

< answer > 24 and 4 </ answer >